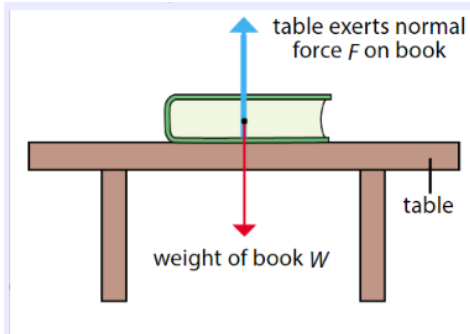


Newton's First Law

if resultant force acting on object = 0, we say forces acting on object are balanced.

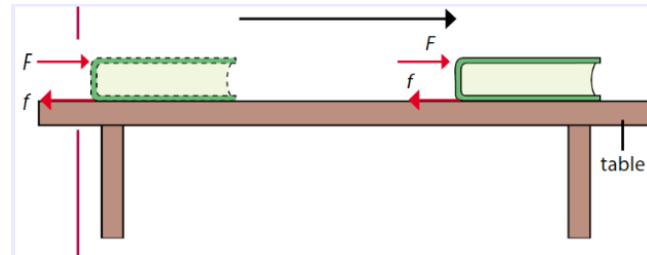
Table exerts an upward force F (normal force) that pushes on the book
 $F = W$ (weight of the book)



- 2 forces acting on book are equal $\Rightarrow$ act in opposite directions.
- thus, resultant force = 0 and the book remains stationary.
- Book is at rest $\Rightarrow$ acceleration = 0

Force F is applied on a book $\Rightarrow$ and it moves in a straight line across a rough table.

$F = f$ (frictional force between book and table)



- resultant force $\Rightarrow$ acting on book are equal $\Rightarrow$ but act in opposite directions
- Resultant force = 0 $\Rightarrow$ book continues moving at constant velocity.

Definition: Newton's first law of motion states that every object will continue in its state of rest or uniform motion in a straight line unless a resultant force acts on it.

#PhysicsDefinitions